

Unit 12, Lesson 3: Multiplying Algebraic Expressions

Benchmarks

MA.B.AR.1.1 Apply the Laws of Exponents to generate equivalent algebraic expressions, limited to integer exponents and monomial bases.

MA.B.AR.1.2 Apply properties of operations to multiply two linear expressions with rational coefficients.

Learning Target

☐ I can apply properties of operations to multiply two algebraic expressions.

12.3.1 Warm-Up: Bell Work

1. Find the product.

$$\frac{3}{2}g^2h^4 \cdot -\frac{5}{2}g^3h$$

2. Fill in the blanks in each expression to make the equation true.

$$\boxed{}x^3y \cdot 2x^2y\boxed{} = -x\boxed{}y^5$$

12.3.2 Guided Practice: Multiplying Monomials by Algebraic Expressions

1. Determine the product of $(-5m^3n^2)$ and $(-2n^2 + \frac{9}{5}n^4)$ using both methods in the table.

Method 1: Use an Area Model	Method 2: Apply the Distributive Property
	$(-5m^3n^2)(-2n^2 + \frac{9}{5}n^4)$

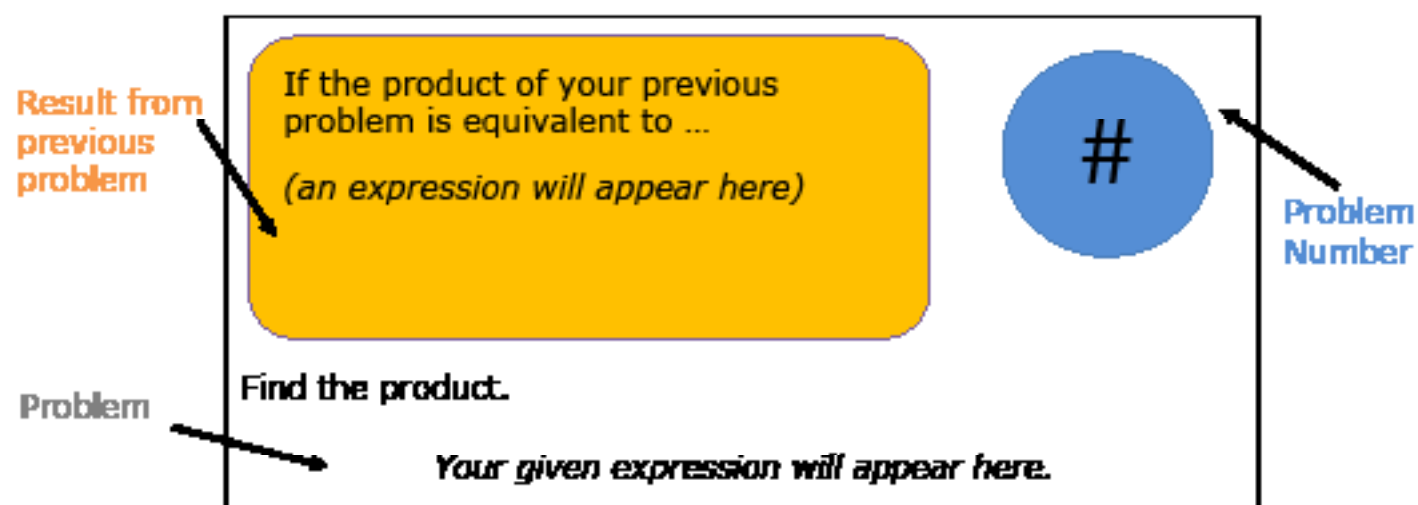
2. Work with your partner to find three expressions equivalent to $3x^2(-2x^3 + x)$.

12.3.3 Collaboration: Equivalent Expressions

1. Two expressions are shown in the table. For each expression, work with your partner to find as many equivalent expressions as you can.
- | Given Expression | $-\frac{3}{2}g^2(g^3 + 2g)$ | $4.2q^2r(-6r^3 - 3.01q)$ |
|------------------------|-----------------------------|--------------------------|
| Equivalent Expressions | | |
2. Find another partner pair and compare your expressions from the previous question. Discuss any differences and add any additional equivalent expressions to your table.

12.3.4 Activity: Scavenger Hunt

Work with your partner or group to complete the scavenger hunt. Each card in the scavenger hunt appears as shown.



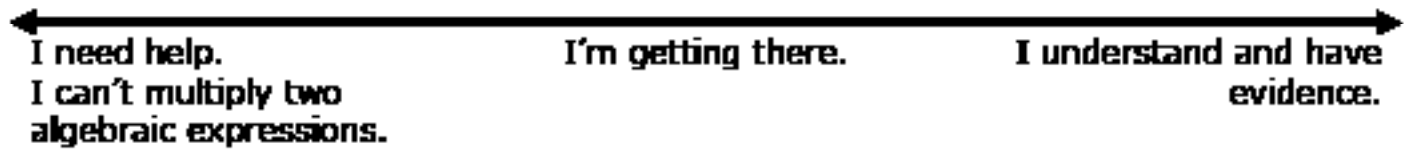
1. Write the number of the problem you start with and the given expression in the first row of the table. Then, work with your group or partner to find the product of the given expression.
2. Look for the card with your product from the previous problem to find your next problem.
3. Repeat this process until your group has completed all the scavenger hunt problems and have ended up back at your original problem.

[illegible]

12.3.5 Wrap-Up: Reflection

1. Draw an **X** on the line to indicate how you feel about today's learning target.

I can apply properties of operations to multiply two algebraic expressions.



2. Explain your placement of "X" on the scale.

